

NanoVDB `ReadAccessor` Benchmark — OLD vs NEW, CPU vs GPU

Benchmarks comparing the `NANOVDB_USE_OLD_ACCESSOR` caching behaviour (ON vs OFF) across access patterns, on CPU (single- and multi-threaded) and GPU.

What is being measured

The `ReadAccessor` caches the tree nodes visited on the previous access so a spatially nearby access can skip the traversal from the root.

- **OLD** (`NANOVDB_USE_OLD_ACCESSOR` defined): on a leaf-cache miss the level-1

and level-2 cache checks are compiled away (`if constexpr` + `else`), so the accessor falls straight to a full **root traversal**.

- **NEW** (`NANOVDB_NO_OLD_ACCESSOR` defined): all applicable cache levels are

always checked, so a leaf-cache miss can still hit the **level-1 / level-2** cache instead of restarting at the root.

Methodology

What the numbers measure

Every reported figure is the **wall-clock time of a single random-access read** into the grid, i.e. one `accessor.getValue(ijk)` call, expressed in **nanoseconds per lookup** (`ns/lookup`). It is a pure *read-path* micro-benchmark: no values are written, no math is done on the result beyond a running sum (kept `volatile` / written to an output array so the compiler cannot elide the work).

- **CPU-1T** — one thread walks the coordinate list serially with a single

accessor. This is a **latency** number: the cost of one dependent lookup.

- **CPU-32T** — the same total work split across 32 threads

(`nanovdb::util::forEach` → `tbb::parallel_for`), each grain using its own accessor. This is a **throughput** number: total time ÷ total lookups with all cores busy.

- **GPU** — the coordinate list is uploaded to the device; each thread processes

a 32-coord chunk with its own accessor. Timed with CUDA events (kernel time only, excluding the one-time grid upload). Also a **throughput** number.

The only thing that differs between the OLD and NEW columns is the accessor's cache-fallback logic (a compile-time switch); the data, coordinates, and harness are identical.

The test data

A **fog-volume sphere** built with `nanovdb::tools::createFogVolumeSphere<float>(radius=500, voxelSize=1.0, halfWidth=3.0)`. Its measured properties:

Property	Value
Value type	float
Index bounding box	<code>[-500, -500, -500] ... [500, 500, 500]</code> (1001 ³ voxels)
Dense voxel count of bbox	1,003,003,001 (~1.00 B)
Active voxels	523,592,077 (~523.6 M)
Fill ratio (active / dense bbox)	52.2 %
Upper internal nodes (level 2, 32 ³)	8
Lower internal nodes (level 1, 16 ³)	272
Leaf nodes (level 0, 8 ³)	82,608
Grid size in memory	188,484,640 bytes (~180 MB)

So it is a **fully populated solid ball** — the interior is all active voxels (the 52 % fill ratio is just the volume of a sphere inside its cube, $\pi/6 \approx 0.524$). It is *sparse* in the VDB sense (only the ball is stored, not the empty corners), but the region the benchmark actually samples is **100 % active**: all access patterns are confined to a cube well inside radius 500, so every lookup lands on a real leaf and traverses the tree for real — we are **not** measuring empty-space shortcuts, we are measuring genuine node traversal and cache reuse.

How much work, how it is averaged

Point patterns (Sequential / LeafJump / NodeJump / Random)	1,048,576 lookups each
Stencil	262,144 centres × 27 neighbours = 7,077,888 lookups
Sampled region	cube of half-extent 256 (points) / <code>[0, 64]</code> ³ dense cube (stencil), both fully inside the ball
Repetition	each configuration run 7 times; the median is reported (robust to OS jitter / turbo ramp)
Warm-up	GPU does one untimed warm-up launch before the 7 timed trials

Hardware / build

CPU	32 hardware threads; TBB <code>parallel_for</code> , 4096-coord grains
GPU	NVIDIA RTX 5000 Ada Generation Laptop GPU (SM 8.9); 32-coord chunks/thread, 128 threads/block
Build	Release (<code>-O3 -DNDEBUG</code>), C++17 / CUDA 17
OLD vs NEW	selected at compile time per target (<code>-DNANOVDDB_USE_OLD_ACCESSOR</code> vs <code>-DNANOVDDB_NO_OLD_ACCESSOR</code>)

Access patterns

Pattern	Stride / shape	Cache behaviour
Sequential	1	leaf (level-0) cache always hot
LeafJump	8 (= leaf dim)	leaf cache cold; level-1 warm (NEW only)
NodeJump	128 (= lower-node dim)	leaf + level-1 cold; level-2 warm (NEW only)
Random	uniform in cube	all caches cold — both fall to root
Stencil	3×3×3 = 27 neighbours per centre, dense sweep	mostly same-leaf hits; boundary neighbours spill to level-1 (NEW rescues)

Combined results — ns per lookup (one `getValue` call), lower = faster

The stencil row is the per-lookup cost inside a 27-neighbour sweep.

Pattern	CPU-1T OLD	CPU-1T NEW	CPU-32T OLD	CPU-32T NEW	GPU OLD	GPU NEW
Sequential (stride 1)	3.57	1.96	0.56	0.46	0.058	0.058
LeafJump (stride 8)	3.82	2.21	0.55	0.49	0.057	0.057
NodeJump (stride 128)	3.94	3.05	0.58	0.53	0.054	0.055
Random (uniform)	11.47	15.32	0.82	1.04	0.088	0.095
Stencil (3×3×3, 27-pt)	4.61	2.13	0.322	0.191	0.086	0.040

Legend: CPU-1T = single thread · CPU-32T = 32 threads (TBB) · GPU = RTX 5000 Ada.

OLD → NEW speedup (>1 = the fix is faster)

Pattern	CPU-1T	CPU-32T	GPU
Sequential	1.82×	1.22×	1.00×
LeafJump	1.73×	1.12×	1.00×
NodeJump	1.29×	1.09×	0.98×
Random	0.76×	0.79×	0.93×
Stencil	2.17×	1.68×	2.17×

Detailed tables

CPU, single-threaded (latency) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	3.57	1.96	1.82×
LeafJump	3.82	2.21	1.73×
NodeJump	3.94	3.05	1.29×
Random	11.47	15.32	0.76×
Stencil	4.61	2.13	2.17×

CPU, 32 threads (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW	MT speedup (NEW)
Sequential	0.56	0.46	1.22×	4.3×
LeafJump	0.55	0.49	1.12×	4.5×
NodeJump	0.58	0.53	1.09×	5.7×
Random	0.82	1.04	0.79×	14.7×
Stencil	0.322	0.191	1.68×	11.1×

GPU (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	0.058	0.058	1.00×
LeafJump	0.057	0.057	1.00×
NodeJump	0.054	0.055	0.98×
Random	0.088	0.095	0.93×
Stencil	0.086	0.040	2.17×

Stencil (3×3×3) — reported per whole 27-neighbour stencil

Platform / Mode	OLD ns/stencil	NEW ns/stencil	OLD → NEW
CPU 1 thread	124.48	57.49	2.17×
CPU 32 threads	8.70	5.17	1.68×
GPU	2.32	1.07	2.17×

Fair platform comparison — NEW accessor, full hardware (ns per lookup)

Workload	CPU-1T	CPU-32T	GPU	GPU vs CPU-32T
Sequential	1.96	0.46	0.058	7.9×
Random	15.32	1.04	0.095	10.9×
Stencil	2.13	0.191	0.040	4.8×

Takeaways

- **Stencil access is the biggest and most universal win.** The realistic finite-difference / convolution pattern gains ~2× from the fix on *both* CPU and GPU. Each centre does 27 correlated lookups; boundary-crossing neighbours that the OLD accessor sends to the root are caught by the NEW accessor's level-1 cache, and the saving compounds across the 27.
- **Coherent point access benefits on CPU** (1.1–1.8×). On GPU isolated point lookups are latency-bound and massively parallel, so the extra cache-level branches are hidden — the fix is neutral (±7%) there.
- **Only pathological fully-random access regresses slightly** (~0.8×), because every cache level misses anyway and the extra checks are pure overhead.
- **The multi-threaded CPU makes the GPU comparison fair.** Against a fully loaded 32-core CPU the GPU is ~5–11× faster (not the misleading 30–170× a single thread would suggest). The gap is smallest for the stencil (4.8×), where heavy cache reuse plays to the CPU's larger per-core caches.
- **CPU multi-thread scaling is memory-bound:** only ~4–7× on 32 cores for coherent patterns (up to ~15× for random / stencil, whose higher per-access latency gives the cores more to overlap).

Building & running

```
# Configure with the benchmark (and CUDA, for the GPU variant) enabled
cmake -S . -B build -DNANOVDDB_BUILD_BENCHMARK=ON -DNANOVDDB_USE_CUDA=ON

# Build all four executables
cmake --build build --target \
    bench_accessor_old bench_accessor_new \
    bench_accessor_cuda_old bench_accessor_cuda_new -j

# Run (CPU prints single- and multi-threaded columns; GPU prints device throughput)
cd build/nanovdb/nanovdb/benchmark
./bench_accessor_old      # CPU, OLD accessor
./bench_accessor_new      # CPU, NEW accessor
./bench_accessor_cuda_old # GPU, OLD accessor
./bench_accessor_cuda_new # GPU, NEW accessor
```

The OLD vs NEW behaviour is selected at compile time per target (`-DNANOVDDB_USE_OLD_ACCESSOR` vs `-DNANOVDDB_NO_OLD_ACCESSOR`), so no reconfigure is needed to compare them.

Files

File	Purpose
BenchPatterns.h	Shared access-pattern generation (identical coords for CPU & GPU)
BenchAccessor.cc	CPU benchmark (single- and multi-threaded)
BenchAccessorCuda.cu	GPU (CUDA) benchmark
CMakeLists.txt	Builds the OLD/NEW × CPU/GPU targets